

**NATURAL SCIENCES TRIPOS    Part II Physics**  
**ADVANCED AND QUANTUM OPTICS**

**Examiners' Solutions and Mark Scheme**

Wednesday 10th June 2026

*Total marks: 56 ( $4 \times 4$  in Section A,  $2 \times 20$  in Section B). Marks shown in square brackets in the right margin. Method marks are awarded for a correct approach even where the final number is wrong; alternative valid derivations should be credited fully.*

**SECTION A**

**Question 1**

*Gaussian beam collimation.*

The Rayleigh range  $z_0$  is the axial distance from the waist over which the beam cross-sectional area doubles; equivalently it is the distance at which the beam radius has grown to  $\sqrt{2}W_0$ , and the distance at which the wavefront curvature is most pronounced. It is given by

$$z_0 = \frac{\pi W_0^2}{\lambda}.$$

[1]

Numerically, with  $W_0 = 5.0 \times 10^{-4}$  m and  $\lambda = 1.0 \times 10^{-6}$  m,

$$z_0 = \frac{\pi(5.0 \times 10^{-4})^2}{1.0 \times 10^{-6}} = \frac{\pi \times 2.5 \times 10^{-7}}{1.0 \times 10^{-6}} \approx 0.785 \text{ m}.$$

[1]

At  $z = 20$  m we have  $z \gg z_0$ , so the beam is in the far field and

$$W(z) = W_0 \sqrt{1 + (z/z_0)^2} \approx W_0 \frac{z}{z_0} = 5.0 \times 10^{-4} \times \frac{20}{0.785} \approx 1.27 \times 10^{-2} \text{ m}.$$

The beam diameter is therefore  $2W(z) \approx 2.5$  cm.

[1]

A larger waist gives a larger Rayleigh range ( $z_0 \propto W_0^2$ ) and a smaller far-field divergence half-angle  $\theta_0 = \lambda/(\pi W_0)$ . The beam therefore stays collimated over a longer distance; physically, a wider beam has a smaller spread of transverse wavevectors and so diffracts less.

[1]

*Marker's note: Accept  $z_0$  defined either via the area doubling or the  $\sqrt{2}W_0$  radius. Full far-field credit if the student uses the exact  $W(z)$  expression and obtains  $\approx 1.27$  cm radius. The final comment must link  $W_0$  to either  $z_0$  or  $\theta_0$ .*

**Question 2**

*Second-order nonlinear processes.*

For a single field  $E(t) = E \cos(\omega t)$  in a medium with a second-order nonlinearity, the nonlinear polarisation is  $P^{(2)} \propto E^2 \cos^2(\omega t) = \frac{1}{2}E^2 [1 + \cos(2\omega t)]$ . The  $\cos(2\omega t)$  term is an oscillating polarisation at twice the input frequency, which radiates a field at  $2\omega$ : this is **second-harmonic generation (frequency doubling)**. The constant term is a static (d.c.) polarisation proportional to the optical intensity: this is **optical rectification**.

[2]

These processes require a non-zero  $\chi^{(2)}$ , which is possible only in a **non-centrosymmetric** crystal (a material lacking inversion symmetry).

[1]

In a centrosymmetric medium, inverting the applied field  $E \rightarrow -E$  must invert the polarisation,  $P \rightarrow -P$ . But a  $\chi^{(2)}E^2$  term is even in  $E$  and so is unchanged under  $E \rightarrow -E$ ; consistency therefore forces  $\chi^{(2)} = 0$ . Equivalently, the anharmonic potential of a centrosymmetric medium has no cubic ( $x^3$ ) term. [1]

*Marker's note: The symmetry argument (parity of  $P$  under  $E \rightarrow -E$ ) is the key point; accept the equivalent statement about the cubic term in the Lorentz potential.*

### Question 3

*Symmetric two-mirror resonator stability.*

For a two-mirror resonator the round-trip ray-transfer matrix gives the stability (confinement) condition

$$0 \leq \left(1 + \frac{d}{R_1}\right) \left(1 + \frac{d}{R_2}\right) \leq 1, \quad \text{i.e.} \quad 0 \leq g_1 g_2 \leq 1.$$

For a symmetric resonator  $R_1 = R_2 = R$ , so  $g_1 = g_2 = g = 1 + d/R$  and the condition becomes  $g^2 \leq 1$ , i.e.

$$-1 \leq 1 + \frac{d}{R} \leq 1 \quad \implies \quad 0 \leq \frac{d}{|R|} \leq 2,$$

where concave mirrors ( $R < 0$ ) are required. [2]

The three special configurations are:

$$\frac{d}{|R|} = 0 \text{ (planar)}, \quad \frac{d}{|R|} = 1 \text{ (confocal)}, \quad \frac{d}{|R|} = 2 \text{ (concentric)}.$$

[1]

The **confocal** resonator,  $d/|R| = 1$ , lies at the centre of the stable range and is the most robust against misalignment. The planar and concentric configurations sit exactly on the stability boundary ( $g_1 g_2 = 1$  or 0), so any small perturbation of mirror tilt or spacing pushes them out of the stable region and rays walk off; the confocal resonator is maximally far from these edges. [1]

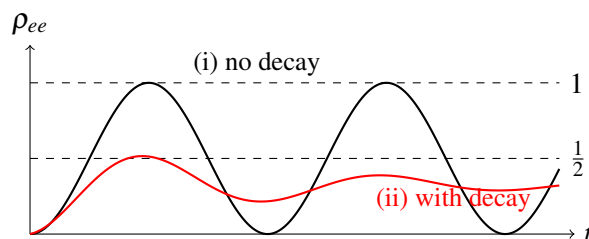
*Marker's note: Derivation of  $0 \leq g_1 g_2 \leq 1$  may be quoted from lectures; the specialisation to the symmetric case and the identification of the three points are required. The robustness argument must refer to confocal being in the middle / planar and concentric being on the boundary.*

### Question 4

*Resonantly driven two-level atom.*

(i) **No spontaneous emission.** Solving the resonant ( $\Delta = 0$ ) two-level equations with the atom initially in the ground state gives  $|c_e(t)|^2 = \sin^2(\Omega_0 t/2) = \frac{1}{2}[1 - \cos(\Omega_0 t)]$ . The excited-state population oscillates undamped between 0 and 1 at the Rabi frequency  $\Omega_0$  — *Rabi oscillations*. At long times it continues to oscillate; there is no steady state. [2]

(ii) **With spontaneous emission**  $\gamma \approx 0.2\Omega_0$ . The optical Bloch equations now include damping. The population still oscillates initially, but the oscillations are damped and the system relaxes to a steady state with  $0 < \rho_{ee} < \frac{1}{2}$ . [1]



Physically, in case (i) the dynamics is fully coherent and energy is periodically exchanged between the atom and the field. In case (ii) each spontaneous-emission event randomises the atomic phase and destroys the coherence; after a few cycles the oscillation washes out and the atom reaches a balance between coherent driving and incoherent decay — the steady state. [1]

*Marker's note: Sketches need not be analytically exact: award the marks for (i) a full-amplitude undamped oscillation reaching 1, and (ii) a damped oscillation settling to a constant value strictly below  $\frac{1}{2}$ . The explanation must mention loss of coherence / approach to steady state.*

## SECTION B

### Question 5

*Optical coherence and coherence-gated imaging.*

- (a) The coherence time is the characteristic “memory time” of the field fluctuations, defined through the power-equivalent width of the normalised temporal coherence,

$$\tau_c = \int_{-\infty}^{\infty} |g(\tau)|^2 d\tau.$$

The coherence length is the corresponding propagation distance,  $\ell_c = c\tau_c$ . The spectral width is the reciprocal of the coherence time,

$$\Delta\nu_c = \frac{1}{\tau_c}.$$

[3]

*Marker's note: Award 1 mark each for  $\tau_c$  (definition or words),  $\ell_c = c\tau_c$ , and  $\Delta\nu_c = 1/\tau_c$ .*

- (b) Writing the two recombining fields as  $U_1 = U(t)$  and  $U_2 = U(t + \tau)$ , both of intensity  $I_0$ , the detected intensity is

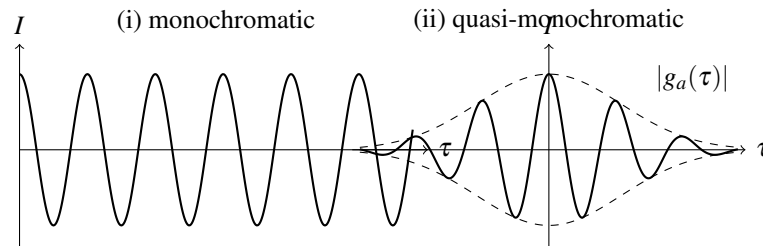
$$I = \langle |U_1 + U_2|^2 \rangle = I_0 + I_0 + 2\text{Re}\langle U^*(t)U(t + \tau) \rangle = 2I_0 [1 + \text{Re}\{g(\tau)\}],$$

using  $g(\tau) = \langle U^*(t)U(t + \tau) \rangle / I_0$ . Writing  $g(\tau) = |g(\tau)| e^{j\varphi(\tau)}$  gives

$$I(\tau) = 2I_0 [1 + |g(\tau)| \cos \varphi(\tau)], \quad \varphi(\tau) = \arg\{g(\tau)\}.$$

[2]

For a perfectly monochromatic wave  $|g(\tau)| = 1$  for all  $\tau$ , so  $I(\tau)$  is a pure cosine of constant unit visibility. For a quasi-monochromatic wave  $|g(\tau)| = |g_a(\tau)|$  decays over  $\sim \tau_c$ , so the fringes have a cosine carrier under an envelope  $|g_a(\tau)|$  that decays away from  $\tau = 0$ : the visibility falls to zero for  $|\tau| \gg \tau_c$ . The envelope of the interferogram therefore directly gives  $|g(\tau)|$  (and the fringe positions give  $\varphi(\tau)$ ). [3]



*Marker's note: The derivation (3 lines) earns 2 marks. The two annotated sketches: 1 mark for the constant-visibility cosine, 1 mark for the decaying envelope, 1 mark for stating that the envelope yields  $|g(\tau)|$ .*

- (c) The coherence length of a source of centre wavelength  $\lambda_0$  and spectral width  $\Delta\lambda$  is, to within a numerical factor of order unity,

$$\ell_c \sim \frac{\lambda_0^2}{\Delta\lambda} = \frac{(800 \times 10^{-9})^2}{25 \times 10^{-9}} = \frac{6.4 \times 10^{-13}}{2.5 \times 10^{-8}} \approx 2.6 \times 10^{-5} \text{ m} \approx 26 \text{ } \mu\text{m}.$$

[2]

A short coherence length is desirable because interference fringes appear only when the two arms are matched to within  $\sim \ell_c$ . The fringe burst from a given reflecting boundary is therefore localised to a narrow range of delay, giving good axial (depth) discrimination and allowing closely spaced layers to be separated — this is *coherence gating*. [2]

*Marker's note: Accept  $\ell_c$  in the range  $\sim 10\text{--}30 \text{ } \mu\text{m}$  depending on the numerical factor used; full marks for  $\lambda_0^2/\Delta\lambda$  with correct arithmetic. The explanation must connect short  $\ell_c$  to depth localisation.*

- (d) Each boundary  $i$  reflects a fraction  $r_i$  of the field with its own delay  $\tau_i$ . The interferogram contains terms  $\propto r_i |g_a(\tau - \tau_i)| \cos[\omega_0(\tau - \tau_i) + \phi_a]$ . Because  $|g_a|$  is narrow (width  $\tau_c$ ), interference fringes appear only when the scanned reference delay  $\tau$  matches  $\tau_i$  to within  $\tau_c$ . Each boundary therefore generates a separate, well-localised fringe burst centred on its own  $\tau_i$ . [2]

The axial separation of two layers is obtained from the difference in the delay positions of their bursts: a delay difference  $\Delta\tau$  corresponds to an optical path difference  $c\Delta\tau$ , hence a physical layer separation  $\Delta z = c\Delta\tau/(2n)$  (the factor of two for the round trip,  $n$  the sample refractive index). [2]

*Marker's note: 1 mark for “fringes only when  $\tau \approx \tau_i$ ”; 1 mark for distinct bursts per boundary; 2 marks for converting the burst-position difference into a physical depth (factor of 2 for round trip expected, but do not penalise heavily if omitted).*

- (e) The axial resolution is set by the coherence length of the source,  $\Delta z \approx \ell_c/(2n) \sim \lambda_0^2/(n\Delta\lambda)$ , which depends only on the optical *spectrum* of the source and not on the focusing geometry. By contrast, in a conventional microscope the axial resolution comes from the depth of field of the focused spot and so scales with the numerical aperture. In OCT the depth information is encoded in the *time delay*, decoupling it from the transverse focusing; hence the axial resolution is independent of NA (only the lateral resolution depends on NA). [4]

*Marker's note: Full marks for clearly stating that OCT axial resolution is fixed by the source spectrum / coherence length whereas microscope axial resolution is fixed by NA-dependent depth of field, with the reason being that depth is encoded in delay. Partial credit (2) for stating the result without the reasoning.*

## Question 6

*The quantised single-mode field and its states.*

- (a) Since  $\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$  and  $\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle$ , both terms in  $\hat{E}_x \propto \hat{a} + \hat{a}^\dagger$  change the photon number by one, so they map  $|n\rangle$  onto states orthogonal to  $|n\rangle$ :

$$\langle n|\hat{E}_x|n\rangle \propto \langle n|(\hat{a} + \hat{a}^\dagger)|n\rangle = \sqrt{n}\langle n|n-1\rangle + \sqrt{n+1}\langle n|n+1\rangle = 0.$$

[2]

For the second moment, with  $\hat{E}_x = \mathcal{E}_0(\hat{a} + \hat{a}^\dagger)$ ,

$$\hat{E}_x^2 = \mathcal{E}_0^2(\hat{a}^2 + \hat{a}^{\dagger 2} + \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger).$$

The  $\hat{a}^2$  and  $\hat{a}^{\dagger 2}$  terms have zero diagonal matrix element. Using  $\hat{a}\hat{a}^\dagger = \hat{a}^\dagger\hat{a} + 1$ ,

$$\langle n|\hat{E}_x^2|n\rangle = \mathcal{E}_0^2\langle n|(2\hat{a}^\dagger\hat{a} + 1)|n\rangle = 2\mathcal{E}_0^2\left(n + \frac{1}{2}\right) \propto \left(n + \frac{1}{2}\right).$$

[2]

A classical field has a well-defined oscillating amplitude, i.e.  $\langle \hat{E}_x \rangle \neq 0$ . A Fock state has  $\langle \hat{E}_x \rangle = 0$  for every  $n$ , no matter how large, even though  $\langle \hat{E}_x^2 \rangle \neq 0$ . The field has a definite “energy” but a completely random phase, so a Fock state does not resemble a classical monochromatic wave. [1]

*Marker’s note: 2 marks for  $\langle \hat{E}_x \rangle = 0$  with the orthogonality argument; 2 marks for  $\langle \hat{E}_x^2 \rangle \propto (n + \frac{1}{2})$  using the commutator; 1 mark for the physical comment about random phase.*

- (b) The probability of finding  $n$  photons is

$$P_n = |\langle n | \alpha \rangle|^2 = \left| e^{-|\alpha|^2/2} \frac{\alpha^n}{\sqrt{n!}} \right|^2 = e^{-|\alpha|^2} \frac{|\alpha|^{2n}}{n!},$$

which is a Poisson distribution with parameter  $|\alpha|^2$ . [2]

The mean photon number is

$$\bar{n} = \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = \langle \alpha | \alpha^* \alpha | \alpha \rangle = |\alpha|^2.$$

[1]

For the variance, use  $\hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} = \hat{a}^\dagger (\hat{a}^\dagger \hat{a} + 1) \hat{a} = \hat{a}^{\dagger 2} \hat{a}^2 + \hat{a}^\dagger \hat{a}$ , so

$$\langle \hat{n}^2 \rangle = \langle \alpha | \hat{a}^{\dagger 2} \hat{a}^2 | \alpha \rangle + \langle \alpha | \hat{a}^\dagger \hat{a} | \alpha \rangle = |\alpha|^4 + |\alpha|^2 = \bar{n}^2 + \bar{n},$$

hence

$$(\Delta n)^2 = \langle \hat{n}^2 \rangle - \bar{n}^2 = \bar{n}.$$

This is the defining signature of a Poisson distribution. [2]

*Marker’s note: 2 marks for showing  $P_n$  is Poissonian; 1 mark for  $\bar{n} = |\alpha|^2$ ; 2 marks for  $(\Delta n)^2 = \bar{n}$  via the normal-ordering step. Accept quoting the variance of a Poisson distribution provided  $P_n$  was first shown to be Poissonian.*

- (c) Because  $\hat{a} | \alpha \rangle = \alpha | \alpha \rangle$  and  $\langle \alpha | \hat{a}^\dagger = \alpha^* \langle \alpha |$ , the field expectation value is non-zero:  $\langle \alpha | (\hat{a} + \hat{a}^\dagger) | \alpha \rangle = \alpha + \alpha^* = 2 \operatorname{Re}(\alpha) \neq 0$ . A coherent state is a superposition of many number states, so the amplitude-changing operators no longer give zero — it has a well-defined mean field with a definite phase. [2]

The fractional uncertainty in photon number is

$$\frac{\Delta n}{\bar{n}} = \frac{\sqrt{\bar{n}}}{\bar{n}} = \frac{1}{\sqrt{\bar{n}}} = \frac{1}{|\alpha|} \rightarrow 0 \quad \text{as } |\alpha| \rightarrow \infty.$$

[1]

Thus for large  $|\alpha|$  a coherent state has both a vanishing relative intensity uncertainty and a well-defined phase, approaching a classical monochromatic wave of fixed amplitude and phase — exactly the idealised output of a laser well above threshold. [1]

*Marker’s note: 2 marks for the non-zero mean field with reasoning; 1 mark for  $\Delta n / \bar{n} = 1 / |\alpha| \rightarrow 0$ ; 1 mark for the link to the classical / laser limit.*

- (d) Acting on the input  $|0\rangle_0 |1\rangle_1 = \hat{a}_1^\dagger |0\rangle_0 |0\rangle_1$  with the given relation,

$$|0\rangle_0 |1\rangle_1 = \hat{a}_1^\dagger |0\rangle = \frac{1}{\sqrt{2}} (i \hat{a}_2^\dagger + \hat{a}_3^\dagger) |0\rangle = \frac{1}{\sqrt{2}} (i |1\rangle_2 |0\rangle_3 + |0\rangle_2 |1\rangle_3).$$

The photon emerges in an entangled superposition of the two output ports, with a 50:50 probability of being detected in either. [2]

The beam-splitter transformation cannot be written consistently using only the populated input mode: the vacuum mode  $\hat{a}_0$  must appear so that the output operators obey the correct commutation relations. Physically, the *vacuum fluctuations* entering the unused port are essential and influence the quantum statistics of the output. **[1]**

*Marker's note: 2 marks for the correct entangled output state; 1 mark for the role of the vacuum port (commutation relations / vacuum fluctuations).*

- (e) In a coincidence measurement *no* coincidences are observed: the two indistinguishable photons always leave together from the same output port (the output state is  $\frac{i}{\sqrt{2}}(|2\rangle_2|0\rangle_3 + |0\rangle_2|2\rangle_3)$ ). This is the Hong–Ou–Mandel effect, arising from destructive interference of the two “both-transmitted” and “both-reflected” amplitudes. **[2]**

By introducing a variable relative delay  $\tau$  between the two photons (e.g. with a delay line) and recording the coincidence rate, one observes a dip — the “HOM dip” — that reaches zero at  $\tau = 0$  and rises as the photons are made distinguishable. The depth and width of the dip quantify the indistinguishability of the photon pair. **[1]**

*Marker's note: 2 marks for “photons leave together / coincidences vanish” with a brief justification (interference of amplitudes). 1 mark for describing the HOM dip as a function of delay as the test of indistinguishability.*

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*End of solutions. Section A: 16 marks. Section B: 40 marks. Paper total: 56 marks.*